

Stoop or Squat Lifting

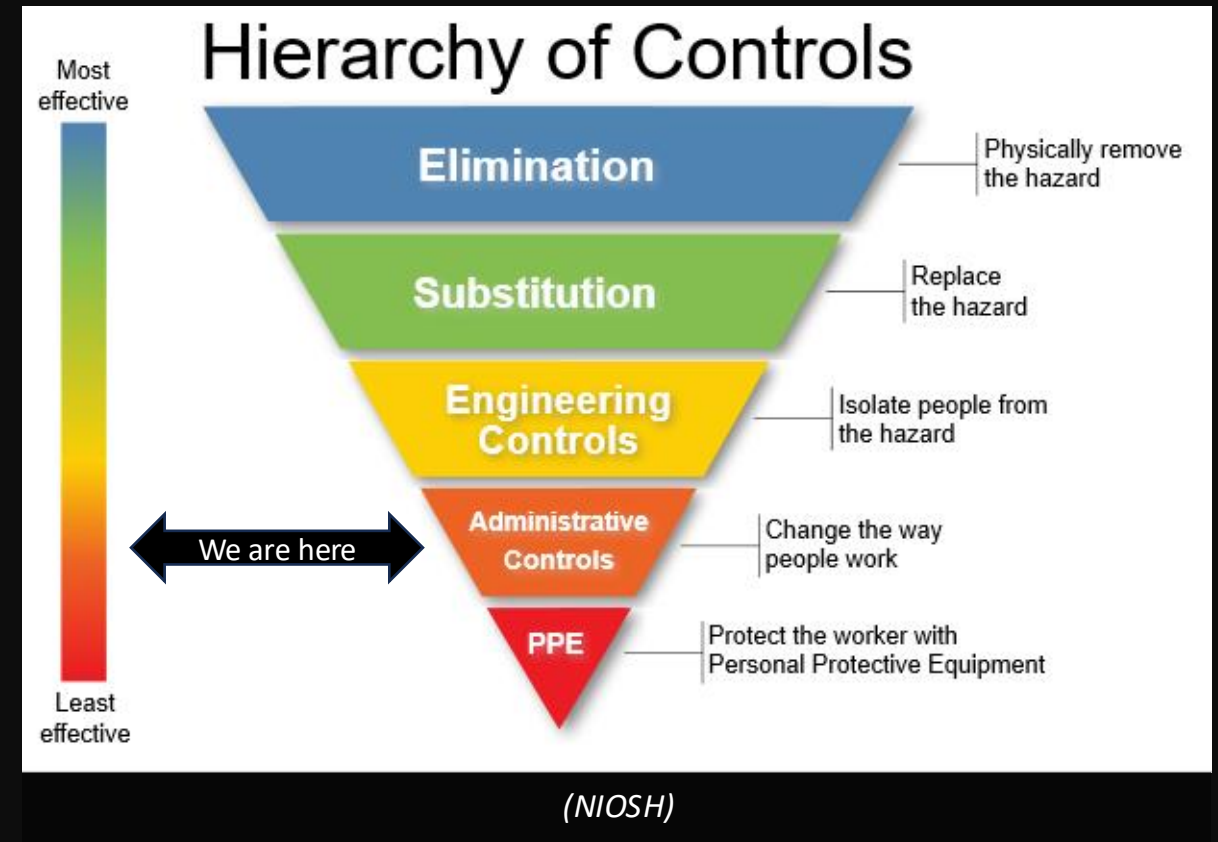
A review of lift technique literature

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ISE 5614

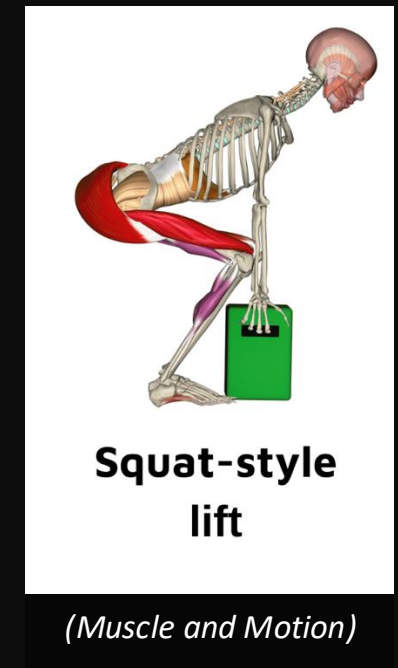
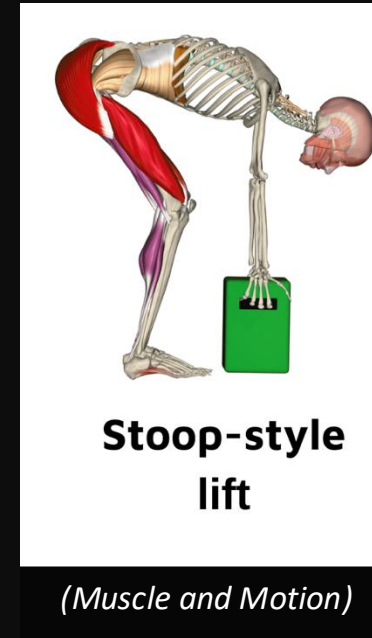
Background

- Low back pain (LBP) is common and costly
 - Lifting is a well-documented risk factor
- Most common method of mitigating is administrative controls
 - Lifting training
- Commonly recommendation is to perform “squat lifts” instead of the more common and comfortable “stoop lift”



Motivation & Objective

- Most lifting technique recommendations support the squat lift
 - Based on the idea that squat lifting reduces the mechanical load on the spine
- Despite being commonly advised and believed, the scientific backing is limited
- Objective
 - To systematically review biomechanical literature comparing the two lift techniques, determining their differences in methodology, and their overall validity



Methodology

- 27 papers from 4 different databases
 - Medline, Current Contents, Embase, and NIOSHTIC
 - Keywords “lifting” and “technique” used
- Assessment Criteria
 - Studies evaluated on methods of measuring or estimating spinal load
 - Intradiscal pressure, spinal shrinkage, net moments, compression, and shear forces
- Validity Rankings
 - Measurement methods and other dependent variables weighted differently on their validity

Table 1

Parameters of back load and their indicators derived from direct measurements or model calculations evaluated for use in comparative studies of lifting technique

Parameter of back load	Indicator	Method	Evaluation ^a	Comments
<i>Measured indicators</i>				
Muscle force	EMG	Surface EMG	0	Low validity
Compression	IDP	Invasive, pressure transducer	3 ★	
	IAP	Radio-pill catheter	0	Inconsistent theory, low validity
	Shrinkage	Stadiometry	1	Low repeatability
Shear	None			
Bending moment + ligament stress	None			
<i>Model based estimates</i>				
Muscle force	Net moment	Static LSM	1	
		Dynamic LSM	2 ✓	
	Estimated extensor moment	Surface EMG, kinematics, muscle length and shortening velocity correction	2 ✓	
Compression + shear	Estimated compression + estimated shear	Net moment, surface EMG	2 ✓	Depends on anatomical fidelity
		Net moment, optimisation	2 ✓	Depends on anatomical fidelity
		Net moment, SEM (constant moment arm)	0 ^b	Little information additional to net moment
Bending moment + ligament stress	Estimated ligament stress, estimated bending moment	Cadaver data, trunk kinematics, force-deformation relationship	2 ✓	

^a 0 = Will be discarded, 1 = will be used in case of converging evidence, 2 = sufficiently valid for comparative use, 3 = valid indicator of back load.

^b Used only if net moments not reported, ranking in these cases according to validity of estimate of net moment.

Table 2
Overview of particulars and main results of the included studies^a

Study	Nr.	Dependent variable	Method	Validity	Subjects	Load	Horizontal position	Stoop	Squat	Diff. % stoop	p-Value	Conclusion	Comments
Andersson et al. (1976)	52	IDP (Mpa)	Direct	3	3 f, 1 m	40	Fixed in front of feet				NS	=	Static pull
Dieën et al. (1994)	74	IDP (Mpa)	Direct	3	3 f, 1 m	10	?	1.83	1.5	18	NS	=	Dynamic lift
		Spinal shrinkage (mm)	Direct	1	11 m	8	25 vs. 30	3.9	4		NS	=	Horizontal position freely chosen
Rabinowitz et al. (1998)	75	Spinal shrinkage (mm)	Direct	1	10 m	20% bm	Fixed in front of feet	4.12	4.81		NS	=	
Park & Chaffin (1974)	76	Net moment (Nm)	Static LSM	1	—	0–16	37.5	81	8.8	−9	—	—	Model study
		Net moment (Nm)	Static LSM	1	—	0–16	50	100	111	−11	—	—	
		Net moment (Nm)	Static LSM	1	—	0–8	75	105	103	2	—	=	
Garg & Herrin (1979)	20	Net moment (Nm)	Static LSM	1	—	0–70	31 vs. 12	256	226	12	—	++	Model study
		Net moment (Nm)	Static LSM	1	—	0–50	38	238	246	−3	—	=	
Ekholm et al. (1982)	77	Net moment (Nm)	Static LSM	1	14 m, 1 f	12.8	Variable, close	217	160	26	<0.001	++	
		Net moment (Nm)	Static LSM	1	14 m, 1 f	12.8	Variable, far	217	200	8	NS	=	
Wax et al. (1987)	78	Net moment (Nm)	Static LSM	1	5 m	25	Variable	182	120	34	<0.001	++	Load between feet in squat
Lindbeck & Arborelius (1991)	79	Net moment (Nm)	Static LSM	1	10 m	12.8	30	186	167	10	<0.01	++	
Mittal & Malik (1991)	80	Net moment (Nm)	Static LSM	1	100 f	18 vs. 17	Variable	295	250	15	<0.05	++	Loads differed between techniques, static holding
Looze et al. (1992)	64	Net moment (Nm)	Static LSM	1	5 m	11	40 ^b	184	185	−1	NS	=	
Bush-Joseph et al. (1988)	81	Net moment (Nm)	Dynamic LSM	2	10 m	15	38			−18	<0.05	—	
Buseck et al. (1988)	82	Net moment (Nm)	Dynamic LSM	2	10 m	5–25	37	309	268	13	?	?	Stoop = free technique, performed faster

Table 2 (continued)

Study	Nr.	Dependent variable	Method	Validity	Subjects	Load	Horizontal position	Stoop	Squat	Diff. % stoop	p-Value	Conclusion	Comments	
Potvin et al. (1991)	83	Net moment (Nm)	Dynamic LSM	2	15 m	6–32	?	290	277	4	?	?		
Lindbeck & Arborelius (1991)	79	Net moment (Nm)	Dynamic LSM	2	10 m	13	30	268	290	−8	NS	=		
Looze et al. (1992)	64	Net moment (Nm)	Dynamic LSM	2	5 m	11	40 ^b	265	255	4	NS	=		
Toussaint et al. (1992)	85	Net moment (Nm)	Dynamic LSM	2	8 m	15	Fixed in front of feet	225 ^b	225 ^b	0	NS	=		
Hagen & Harms-Ringdahl (1994)	86	Net moment (Nm)	Semi-dynamic LSM	2	10 m	1–17	Fixed in front of feet	200 ^b	193 ^b	0	NS	=		
Dolan et al. (1994a)	87	Extensor moment (Nm)	EMG based	2	23 m, 126 f	10	?	243	252	−4	<0.05	−	👎	
Dolan et al. (1994b)	88	Extensor moment (Nm)	EMG based	2	21 m, 18 f	0–30	?	228	254	−11	<0.0001	--	👎	
Dieën et al. (1994)	74	Net moment (Nm)	Dynamic LSM	2	11 m	8	25 vs. 30	196	214	−9	0.04	−	👎	Horizontal position freely chosen
Toussaint et al. (1997)	89	Net moment (Nm)	Dynamic LSM	2	8 m	20% BM	51.5 ^b	368	366	1	NS	=		
Kjellberg et al (1998)	94	Net moment (Nm)	Dynamic LSM	2	12 f	8.7	50 ^b	150	151	−1	NS	=		
Looze et al. (1998)	90	Net moment (Nm)	Dynamic LSM	2	8 m	7–16	38 ^b	210	228	−9	<0.001	--	👎	
Troup et al. (1983)	91	Compression (N)	Dynamic LSM + SEM	2	10 m	15	In front of feet	5767	6039	−5	<0.001	−	👎	Feet placed asymmetrically in squat
Leskinen et al. (1983)	92	Compression (N)	Dynamic LSM + SEM	2	20 m	15	Fixed feet under load	6365	5866	8	<0.01	+	👍	
Leskinen et al. (1985)	63	Compression (N)	Static LSM + SEM	1	20 m	15	Fixed feet under load	3989	4033	−1	NS	?		
Anderson & Chaffin (1986)	73	Compression (N)	Static LSM + opt.	1	1	5–40	Between legs	5055	3304 ^c	35	−	++	👍	Crude anatomical model
		Compression (N)	Static LSM + opt.	1	1	5–40	In front of feet	3931	4275 ^c	−9	−	−	👎	Partly predicted kinematics
Potvin et al. (1991)	83	Compression (N)	Dynamic LSG + EMG	2	15 m	15	?	4275	4724	−11	<0.05	--	👎	
Potvin et al. (1992)	84	Compression (N)	Dynamic LSM + EMG	2	13 m	6–32	34 vs. 32			−6 ^b	0.02	−	👎	

Table 2 (continued)

Study	Nr.	Dependent variable	Method	Validity	Subjects	Load	Horizontal position	Stoop	Squat	Diff. % stoop	p-Value	Conclusion	Comments
Chaffin & Page (1994)	93	Mass at 3400N comp. (Kg)	Static LSM + opt.	1	—	Dep. var.	38-58	16 ^b	10 ^b		—	-- 🚫	Model study, predicted kinematics
Potvin et al. (1991)	83	Shear (N)	Dynamic LSM + EMG	2	15 m	15	?	450	156	65	<0.05	++ 👍	
Anderson & Chaffin (1986)	73	Fascial strain (%)	Kinematics based	2	1	5–40	Between legs	3.6	2.3 ^c	36	—	++ 👍	Partly predicted kinematics
		Fascial strain (%)	Kinematics based	2	1	5–40	In front of feet	8.8	2.4 ^c	73	—	++ 👍	Partly predicted kinematics
Dolan et al. (1994a)	87	Bending moment (Nm)	Kinematics based	2	23 m, 126 f	10	?	75	54	28	<0.05	++ 👍	
Dolan et al. (1994b)	88	Bending moment (% max)	Kinematics based	2	21 m, 18 f	0–30	?	31	18	42	<0.05	++ 👍	

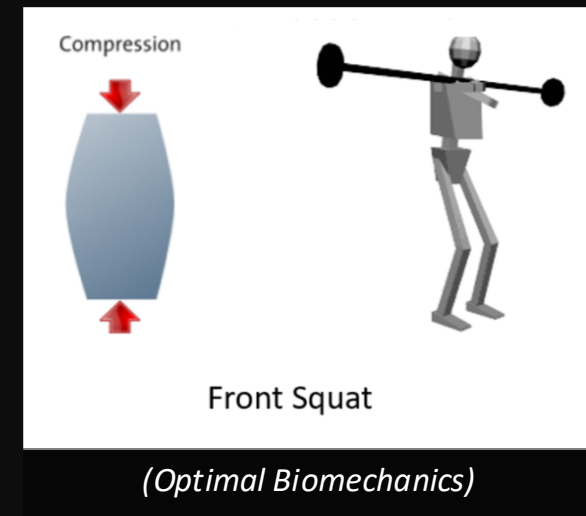
^a m – male; f – female; -- – substantial (>10%) negative effect of squat technique; – is limited (<10%) negative effect of squat technique; = – no effect of lifting technique; + – limited (<10%) positive effect of squat technique; ++ – substantial (>10%) positive effect of squat technique.

^b Estimated.

^c Curved back squat and flat back squat averaged.

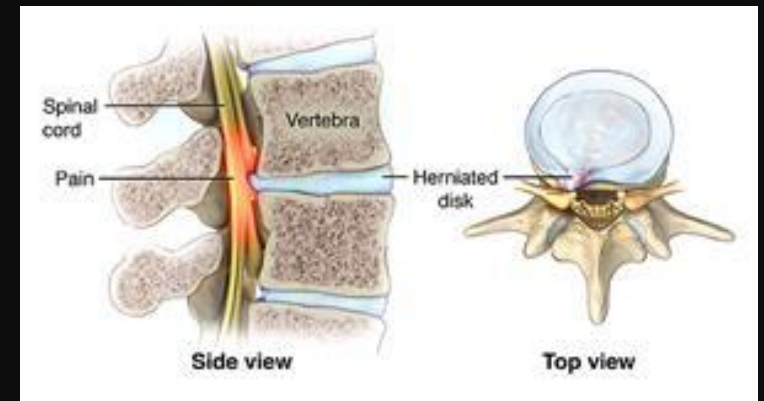
Spinal Compression & Shrinkage

- Studies found no significant differences in spinal compression between stoop and squat lifting
 - Indicated by intradiscal pressure and spinal shrinkage
- Both lifting techniques produced compression forces that could cause injury
 - More likely if repeated frequently or if performed by vulnerable populations



Net Moments & Compression

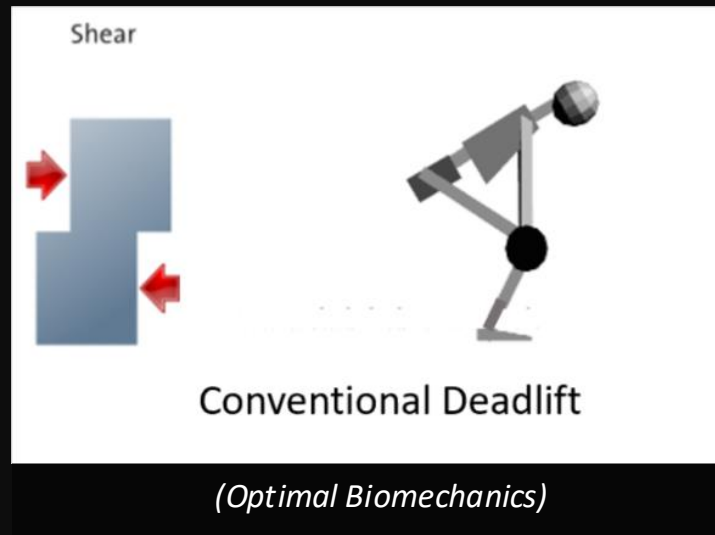
- Model estimates showed either no difference or slightly higher loads in squat lifting
 - One exception to this; when the load can be placed between the feet
- Dynamic models show higher vertical ground reaction forces due to greater COM vertical movement



(Johns Hopkins)

Shear & Bending Forces

- Shear and bending forces were consistently lower in squat lifting, suggesting a potential benefit to specific spinal stresses
 - Reducing the risk of disks slipping from overexertion



Energy Costs & Fatigue

- Multiple papers mentioned energy costs and fatigue
- The squat technique was associated with higher energy costs and faster fatigue onset
 - Likely causing a reduction in compliance with “proper” lifting technique in the workplace

Conclusions

- While squat lifting reduces spinal shear and bending forces, this is not sufficient evidence to justify its universal recommendation given the lack of significant differences in compression forces and the practical challenges associated.
 - Worker compliance, fatigue
- Benefits of squat lifting are context-specific, limiting its utility
 - If the load can be placed between the feet or not
- More important to preventing LBP than lift technique are factors such as lifting speed, load positioning, and task asymmetry.
 - Factors that all play into the NIOSH Lifting Equation

Implications

- Challenges the common misconception that squat lifting is universally effective in preventing LBP
- Suggests LBP prevention should revolve around more impactful changes
 - Engineering controls
 - Ergonomic workplace adjustments

Reference

[1] Jaap H. van Dieën, Marco J.M. Hoozemans, and Huub M. Toussaint. "Stoop or squat: a review of biomechanical studies on lifting technique." *Clinical biomechanics* 14.10 (1999): 685-696.

Thank you!

Any questions?